Supplementary Tables 6. The inclusion criteria in each selected study

1.EMMA

- Younger than 75 years;
- Diastolic office blood pressure readings ＞95 mm Hg on at least three occasions and/or receiving antihypertensive treatment；
- Creatinine clearance of≥0.83 mL/s (50 mL/min);
- Anatomic inclusion criteria：
    i. The atherosclerotic nature of the RAS, as inferred from renal artery and aortic views;
    ii. A reduction in arterial diameter of either ＞75% without thrombosis or of ＞ 60% with a positive lateralization test (lateralized intravenous pyelography, renal scintigraphy, or renal vein renin determination, performed according to the usual practice of each center);
    iii. A stenosis affecting the main renal artery, which had not been previously dilated;
    iv. A functional kidney on the opposite side exhibiting a normal main artery or an arterial diameter reduction ＜50%.

2. SNRASCG

- Sustained hypertension, with a minimum diastolic BP of 95 mm Hg on treatment with at least two antihypertensive drugs;
- Bilateral (stenosis of 50% or greater in arteries to both kidneys or in the artery to a solitary functioning kidney);
- Unilateral (stenosis of 50% or greater in one renal artery and less than 50% stenosis on the contralateral side)

3. DRASTIC

- Renal function (defined as a serum creatinine concentration of «2.3 mg per deciliter [200 μmol per liter]);
- Patients receiving standard antihypertensive medication in whom diastolic blood pressure remained > or =95 mmHg during three consecutive visits to the outpatient clinic underwent full diagnostic work-up, including renal arteriography;
- Unilateral or bilateral renal-artery stenosis of at least 50 percent.

4. STAR

- Age >18 years
- Ostial atherosclerotic renal artery stenosis ≥50% on CTA, MRA or intra-arterial angiography
- Estimated creatinine clearance <80 ml/min/1.73m2 according to the Cockcroft and Gault formula, on two occasions within one month

5. RASCAD

- Ischemic heart disease
- Angiographic diagnosis of atherosclerotic RAS >50% and ≤80%

6. CORAL

- Either:

a. Documented history of hypertension on 2 or more anti-hypertensive medications OR

b. Renal dysfunction defined as Stage 3 or greater CKD based on the new National Kidney Foundation classifications (estimated GFR < 60 mL per minute per 1.73 m2 calculated by the modified MDRD formula)

- One or more severe renal artery stenoses by any of the following pathways:
  a. Angiographic: ≥ 60% and < 100% by renal angiogram OR
  b. Duplex: systolic velocity of >300 cm/sec OR
  c. Core lab approved Magnetic Resonance Angiography (MRA) demonstrating:
    - Stenosis > 80% OR
    - Stenosis > 70% with spin dephasing on 3D phase contrast MRA OR
    - Stenosis > 70% and two of the following:
      i. Ischemic kidney is > 1 cm smaller than contralateral kidney.
      ii. Ischemic kidney enhances less on arterial phase.
      iii. Ischemic kidney has delayed gadolinium excretion.
      iv. Ischemic kidney hyper-concentrates the urine.
      v. 2-D phase contrast flow waveform shows delayed systolic peak
      vi. Post-stenotic dilatation
  d. Core lab approved Computerized Tomographic Angiography (CTA) demonstrating:
    - Stenosis is > 80% on high quality CTA.
    - Stenosis is > 70% on CTA and there are two of the following:
      i. The length of ischemic kidney is > 1 cm smaller than contralateral kidney.

ii. Reduced cortical thickness of ischemic kidney.
iii. Less cortical enhancement of ischemic kidney on arterial phase.
iv. Post-stenotic dilatation

7. RADAR

- Age ≥ 18 years
- Informed consent signed by patient (and/or legal guardian),
- Hemodynamically relevant de novo unilateral or bilateral atherosclerotic renal artery diameter stenosis RAS (≥ 70%)
- Estimated GFR > 10 ml/min calculated using the abbreviated Modification of Diet in Renal Disease (MDRD) Study equation,
- Patients presenting mild, moderate or severe hypertension (defined according to the WHO guidelines) and/or renal dysfunction,
- Target lesion must be completely coverable by one study stent,
- Total target lesion length estimated to be less than 19 mm,
- Target lesion accessible to direct stenting or, after pre-dilation, is likely to sufficiently benefit from stenting (at the discretion of the investigator),
- Renal reference vessel diameter (RVD) of ≥ 4.0 mm and < 7.0 mm based on visual estimation,
- Willingness to comply with all the specified follow-up evaluations.

8. NITER

- Age ≤ 80 years
- Presence of ostial renal artery stenosis ≥ 70% (determined by Duplex Doppler ultrasonography and confirmed by Magnetic Resonance valuated by at least two experienced operators)
- Serum creatinine ≤ 3 mg/dl and/or creatinine clearance (Modification of Diet in Renal Disease, MDRD formula)≥ 30 ml/min, defined as stage 3 or greater CKD based on National Kidney Foundation classification
- Longitudinal ultrasonographic diameter of the stenotic kidney ≥ 8 cm
- Blood pressure values ≤ 150/90 mmHg with the use of less than four hypotensives drugs

9. ASTRAL

- Atherosclerotic renovascular disease (ARVD) confirmed angiographically
- At least one ARVD lesion that is suitable for revascularisation

- No definite indication for, or contraindication to, revascularisation, and revascularisation unlikely to become definitely indicated within 6 months of entry
- Informed consent obtained from patient